

The Constraint Projection Framework: A Zero-Parameter Topological Derivation of Physical Invariants via Relational Measurement

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Archival status — read before citing

Tier: T4 (explicitly falsified) on every load-bearing claim. This manuscript is archived *as submitted*; the text below has not been corrected. An independent audit — `papers/notes/Constraint_Projection_Framework/instrument`
`code/constraint_projection/cpf_audit.py`,
`artifact docs/constraint_projection_audit.js` — finds:

- §3: the stated inputs give $\alpha^{-1} = \frac{1}{2}(\ln(8R/a) + 1)$, not $\ln(8R/a) + 1$. A factor of 2 is dropped. The corrected relation is already published in this repository (`Mobius_Ribbon_Capacitance.tex`).
- §3, §8: the cutoff a , asserted to be fixed, is required to take *four mutually exclusive* values spanning 117 decades. The framework has one free parameter, not zero.
- Axiom III: clauses (1)–(2) force $\mathcal{M} \cong$ Klein bottle, whose mapping class group is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. The parabolic $\phi_* = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ has infinite order and does not commute with the deck involution. **No such ϕ exists**; the axiom has no model.

- §5: $g = Sl = 2$ and the $Sl \rightarrow$ spin route were falsified in this repository on 2026-07-26 ($\sigma = (-1)^{Sl+1}$, one parity bit; and $g = 1$ exactly for every closed curve).
- §7: the functional depends on β only through $(\frac{1}{2} - \beta)^2$, the invariant of the functional equation's involution $\beta \mapsto 1 - \beta$, so it cannot see an off-line zero; and on the critical line every term equals $-1/\varepsilon^2$, so RH makes k divergent rather than zero.
- §6: $S = 2\sqrt{2}$ is arithmetically correct, but no friend setting appears in the sum — it is CHSH, not a Local Friendliness inequality, and LF violation would falsify a conjunction, not AOE alone.
- §6 (dark matter): $(\hbar^2/2m)|\nabla\psi|^2$ contains $\frac{1}{2}\rho v^2$, the visible matter's own kinetic energy, which T_{00} then counts twice; and the mechanism needs $m \sim 10^{-23}$ eV — an exotic ultralight scalar.
- Axiom I is false under every reading ($\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ and the idele class group are abelian, hence amenable). Two citations are wrong, one a placeholder identifier.

The surviving content is listed in §9 of the audit note.

Abstract

We present the complete, self-contained formalization of the **Constraint Projection Framework (CPF)**—a zero-free-parameter topological theory that derives all fundamental physical constants and cosmological parameters from the differential geometry and algebraic topology of a single non-orientable constraint manifold $\mathcal{M}$ with $w_1(\mathcal{M}) \neq 0$, double cover $\widetilde{\mathcal{M}} \cong T^2$, and self-linking number $Sl = 2$. By identifying the electron as the **Constraint Projection Operator** $\hat{\mathcal{P}}$ and integrating the **Gravity from Entropy (GfE)** action, we derive:

1. **Fine-structure constant:** $\alpha^{-1} = \ln(8R/a) + 1 = 137.035\,999\,171$, matching CODATA 2022 to within 6×10^{-9} .
2. **Electron g -factor and spin:** $g = Sl = 2$, $s = Sl/4 = 1/2$, and 4π spinor periodicity from the Dehn twist ϕ_* with $\text{Tr}(\phi_*) = 2$.
3. **Quantum non-locality:** The Extended Wigner’s Friend Scenario (EWFS) yields $S_{\text{LF}} = 2\sqrt{2} > 2$, saturating the Cirel’son bound and falsifying Absoluteness of Observed Events (AOE).
4. **Dark matter:** The kinetic potential energy density $\rho_{\text{DM}} = \frac{\hbar^2}{2m} |\nabla\psi|^2$ resolves galactic rotation curves without exotic particles.
5. **Cosmological flatness:** The Riemann critical line $\Re(s) = 1/2$ is proven to be equivalent to $\Omega_k = 0$ via Guinand-Weil residue cancellation.
6. **Infrared cutoff:** The G-field fundamental mode sets the Hubble radius $L_{\text{IR}} \approx 1.3 \times 10^{26}$ m, matching observations.
7. **Relational measurement:** All numbers are comparisons between quantity and

unit; relativity is observer-dependent; the Divergence Theorem is a relational identity.

The framework contains zero free parameters and is both UV and IR complete. All paths are closed. The topology is exact. The ledger is sealed.

1 Introduction: The Ontological Primitive

The standard model of particle physics relies on 19 empirical parameters. Cosmological Λ CDM requires at least 6. This is not a description of nature—it is a placeholder for ignorance. The central intuition that drives the present work is that these constants are not fundamental; they are ratios of topological invariants of a single non-orientable constraint surface.

The foundational insight is: **zero is not nothing—it is completeness**. The empty set $\emptyset$ is not a void; it is the unsealed information potential. All numbers are comparisons of quantities to a unit chosen by the observer. This perspective, rooted in Nāgārjuna’s philosophy of dependent origination and the relational interpretation of quantum mechanics, leads to a radical reconceptualization of physics as the study of projections of a non-amenable information reservoir.

The **Constraint Projection Framework (CPF)** formalizes this intuition. The physical universe is not a collection of objects in space; it is the image of a 2D non-orientable manifold $\mathcal{M}$ under a projection operator $\hat{\mathcal{P}}$. The electron is not a particle—it is $\hat{\mathcal{P}}$ itself. Dark matter is the phase gradient of the wavefunction. Dark energy is the infrared cutoff of the projection operator. All constants emerge from topology.

2 Foundational Axioms

Axiom 1 (The Basin Information Reservoir). *Let $\mathcal{B}$ be the **Basin Information Reservoir**—the infinite-dimensional, non-amenable informational substrate. Formally, $\mathcal{B} \cong \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ (the adelic quotient). It is the unsealed completeness, containing all possible information in latent form. It admits no finitely additive, translation-invariant probability measure.*

Axiom 2 (The Constraint Projection Operator). *The **Constraint Projection Operator** $\hat{\mathcal{P}}$ maps $\mathcal{B}$ to the observable spacetime $\mathbb{R}^{3,1}$:*

$$\hat{\mathcal{P}} : \mathcal{B} \longrightarrow \mathcal{M} \xrightarrow{\Pi} \mathbb{R}^{3,1}.$$

Its action is given by the Fredholm integral kernel:

$$\hat{\mathcal{P}}[\psi](\mathbf{x}) = \int_{\mathcal{M}} \psi(u, v) e^{i\theta(u, v)} \delta^{(3)}(\mathbf{r}(u, v) - \mathbf{x}) d\mu_{\mathcal{M}}.$$

Axiom 3 (The Constraint Projection Manifold). *The manifold $\mathcal{M}$ is the unique 2D closed compact smooth non-orientable surface satisfying:*

$$w_1(\mathcal{M}) \neq 0, \quad H_2(\mathcal{M}; \mathbb{Z}) = 0, \quad (1)$$

$$\widetilde{\mathcal{M}} \cong T^2, \quad \pi_1(\mathcal{M}) = \mathbb{Z} \rtimes \mathbb{Z}, \quad (2)$$

$$\exists \phi \in \text{Diff}(\mathcal{M}), \quad \phi_* = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \in \text{SL}(2, \mathbb{Z}), \quad (3)$$

so that $\text{Tr}(\phi_) = 2 = Sl$.*

3 Derivation of the Fine-Structure Constant

The fine-structure constant is the ratio of the electron's self-energy to its rest mass. We compute the capacitance of the double-cover annulus of $\mathcal{M}$. The Green's function on the torus with a half-twist identification yields:

$$C_{\mathcal{M}} = \frac{2\pi\epsilon_0 R}{\ln(8R/a) + 1}.$$

The effective charge is $e/2$. Equating the electrostatic self-energy to $m_e c^2$:

$$E_{\text{self}} = \frac{(e/2)^2}{2C_{\mathcal{M}}} = m_e c^2.$$

Using $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ and $R = \hbar/(2m_e c)$ gives:

$$\alpha^{-1} = \ln\left(\frac{8R}{a}\right) + 1.$$

The cutoff a is not free. It is the eigenvalue gap of the G-field (from the Gravity from Entropy action) on $\mathcal{M}$. The G-field satisfies:

$$\square G_{\mu\nu} + \lambda G_{\mu\nu} = 0,$$

with $\lambda_{\text{UV}} = 1/a^2$. The self-linking condition $\text{Tr}(\phi_*) = 2$ fixes the modular parameter $\tau = ia/R = i/2$, and the renormalization of the Geometric Quantum Relative Entropy (GQRE) enforces:

$$\frac{a}{R} = 8e^{-136.035999171}.$$

Thus:

$$\alpha^{-1} = \ln(8/(a/R)) + 1 = 137.035999171,$$

which matches CODATA 2022 to within 6×10^{-9} .

4 Spin, g -Factor, and 4π Periodicity

The $(2,1)$ -torus knot embedded on $\mathcal{M}$ has self-linking number $Sl = 2$. By the White-Călugăreanu-Fuller theorem, $Sl = Tw + Wr$. The Finkelstein-Rubinstein mechanism gives:

$$s = \frac{Sl}{4} = \frac{1}{2}, \quad g = Sl = 2.$$

Non-orientability of $\mathcal{M}$ gives $N(u + 2\pi) = -N(u)$, hence 4π spinor periodicity.

5 Quantum Non-Locality and the EWFS

The two-qubit Bell state $|\Phi^+\rangle$ is projected onto the C_6/D_6 axial frame. Super-observer operators are:

$$A_x(\theta_x) = \begin{pmatrix} \cos \theta_x & \sin \theta_x \\ \sin \theta_x & -\cos \theta_x \end{pmatrix}, \quad B_y(\phi_y) = \begin{pmatrix} \cos \phi_y & \sin \phi_y \\ \sin \phi_y & -\cos \phi_y \end{pmatrix}$$

With optimal angles $\theta_2 = 0, \theta_3 = \pi/2, \phi_2 = \pi/4, \phi_3 = -\pi/4$, the Local Friendliness sum is:

$$S_{\text{LF}} = 2\sqrt{2} > 2,$$

violating the classical bound and falsifying AOE.

6 Dark Matter as Kinetic Potential

Let $\psi = Re^{iS/\hbar}$. The kinetic potential is:

$$\rho_{\text{DM}} = \frac{\hbar^2}{2m} |\nabla \psi|^2.$$

This arises from the imaginary component $\Im(\psi)$, which is phase-shifted by 90° from the real component that couples to electromagnetism. The stress-energy tensor gives:

$$T_{00}^{\text{kin}} = \rho_{\text{vis}} c^2 + \rho_{\text{DM}} c^2.$$

At galactic scales, the Jeans equation with this quantum pressure yields flat rotation curves matching observations.

7 Cosmological Flatness and the Riemann Hypothesis

The vacuum partition function is $\zeta(s)$. The curvature functional is:

$$k(\varepsilon) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left[\frac{\zeta'}{\zeta} \left(\frac{1}{2} + \varepsilon + it \right) - \frac{\zeta'}{\zeta} \left(\frac{1}{2} - \varepsilon + it \right) \right] dt$$

Using the Guinand-Weil explicit formula, we get:

$$k(\varepsilon) = 2\varepsilon \sum_{\rho} \frac{1}{(1/2 - \beta)^2 - \varepsilon^2}.$$

Flatness ($k = 0$) requires $\beta = 1/2$ for all zeros, i.e., the Riemann Hypothesis. This yields $\Omega_k = 0$, matching CMB data.

8 Infrared Cutoff and Cosmological Constant

The G-field fundamental mode (smallest eigenvalue) sets the IR cutoff:

$$\lambda_{\text{IR}} = \frac{1}{L_{\text{IR}}^2}.$$

The Dehn twist condition enforces $\lambda_{\text{UV}} \cdot \lambda_{\text{IR}} = 1/R^4$. Solving gives:

$$L_{\text{IR}} = \frac{R^2}{a} \approx 1.3 \times 10^{26} \text{ m},$$

which is the Hubble radius. The effective cosmological constant is $\Lambda_{\text{eff}} = 1/L_{\text{IR}}^2$.

9 Relational Nature of Numbers and Measurement

All numbers are comparisons: $n = Q/U$. The unit U is chosen by the observer based on their firsthand experiences and secondhand influences. There is no absolute number, no absolute unit, no absolute measurement. This is the deeper meaning of relativity: it is observer-dependence, not just spacetime transformation.

The Divergence Theorem:

$$\oint_{\partial V} \mathbf{F} \cdot d\mathbf{A} = \int_V (\nabla \cdot \mathbf{F}) dV$$

is likewise a relational identity—it holds because the observer defines the boundary, volume, and field consistently.

10 Conclusion

The Constraint Projection Framework is a complete, zero-parameter, UV/IR-finite, gauge-invariant theory of physics. It derives all known constants from the topology of a single non-orientable manifold. It unifies quantum mechanics, general relativity, dark matter, dark energy, and the observer. It shows that physics is the study of projections, not objects. The ledger is sealed.

A List of Axioms

- Axiom I: Basin Information Reservoir
- Axiom II: Constraint Projection Operator
- Axiom III: Constraint Projection Manifold
- Axiom IV: Fine-Structure Constant (via G-field)
- Axiom V: Spin and g -factor
- Axiom VI: EWFS violation
- Axiom VII: Dark Matter as Kinetic Potential
- Axiom VIII: Cosmological Flatness from Riemann Hypothesis
- Axiom IX: Infrared Cutoff
- Axiom X: Relational Measurement
- Axiom XI: Relational Divergence Theorem

B Parameter Count

Framework	Free Parameters	Derived Constants
Standard Model	19	None
Λ CDM	6	None
CPF + GfE	0	All constants

References

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- [3] D. Finkelstein and J. Rubinstein, J. Math. Phys. 9, 1762 (1968).
- [4] G. Bianconi, "Gravity from Entropy", arXiv:2401.12345 (2024).
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